

Date: 28.11.2012, Wednesday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1 Do as Directed.

10

- I. _____ layer is responsible for translating into data frames.
- II. Transport layer appends port address. [True/False]
- III. FTP works at the Application layer of the TCP/IP model. [True/False]
- IV. Switch is a device that transmits data by following unicast transmission. [True/False]
- V. Which address is used to locate application running on remote machine?
- VI. Define multiplexing.
- VII. _____ is connection less protocol at transport layer.
- VIII. Calculate the sampling rate for a signal having frequency varies from 0 to 10Khz.
- IX. Loss of signal strength during transmission is known as attenuation. [True/False].
- X. Define Encoding.

Q-2

- a) List the responsibilities of Session Layer. Briefly explain each. 5
 - b) What is protocol? Explain key elements of a protocol. 3
 - c) Explain Amplitude Shift Keying 5
 - d) Multiplexing and de multiplexing is also applicable for the TCP/IP protocol. 2
- [True/False]. Justify.

OR

Q-2

- a) List the techniques to perform analog to digital modulation. Explain pulse code modulation. 5
- b) Explain Amplitude modulation with necessary diagram. 5
- c) What is routing table? Discuss various fields of routing table in brief. 5

- Q-3 Attempt any TWO.** **10**
- Discuss a situation in which time division multiplexing works more efficiently than frequency division multiplexing. Explain with necessary sketch.
 - Define following i) wavelength ii) throughput iii) amplitude iv) propagation delay v) distortion.
 - Discuss transmission modes.

SECTION-II

- Q-4**
- Write a short note on external modem. **4**
 - Write a difference between following: **4**
 - FDM Vs TDM.
 - Asynchronous TDM Vs Synchronous TDM.
 - Compare TCP Vs UDP. **4**
- Q-5**
- Calculate CRC for 10 bit data 1010011110 and divisor is 1011. **4**
 - Draw the signal pattern by following Manchester encoding technique for the bit stream 10001001111. **5**
 - What do you mean by polar encoding? Explain NRZ-L and NRZ-I with necessary sketch. **5**

OR

- Q-5**
- Gateway works at which layer in TCP/IP protocol stack? justify. **4**
 - Why we should go with the dynamic routing for the internetwork? Compare periodic updates Vs triggered updates in the context of dynamic routing process. **5**
 - Compare OSI reference model Vs TCP/IP protocol stack. **5**
- Q-6 Attempt any THREE.** **9**
- List out various components of Data Communication system.
 - Discuss in brief Go back-N technique.
 - Define noise and discuss its implications on analog signal and digital signal.
 - Discuss LRC technique with appropriate example.